CBAM engine: formulas, routes, tooling and what changed

Technical reference · 8 August 2026 · Delta Climate Research

Every figure below was read from the pack live at `deltaclimate.earth` on the date of writing, not from a local build. Where something is unverified it says so.

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1. What shipped

Four changes, across two repositories, all live.

Commit	Repo	What
`397457e`	CBM	Production-year markers become validity windows; generator emits its own thresholds again
`95224d0`	CBM	Real OJ hashes pinned for IR 2025/2620 and 2025/2621
`d529b1b`	Angad	Re-vendor: 185 unpriceable goods become 23
`bf211f1`	Angad	Re-vendor: the two regulations now carry real hashes

Effect on the calculator, measured across all 574 catalogue CN codes at five origins, import date in 2026:

	Before	After
Goods that produce an estimate	385	**547**
Goods that cannot be priced from any origin	185	**23**

No calculated value changed. A migration gate proved the multiset of `(CN code, column, benchmark value)` identical before and after. The recovered goods were previously returning `status: 'unavailable'` — the engine refused rather than mispriced. This is coverage, not accuracy.

Also this branch: multi-line, a real threshold verdict, and export

Six more changes, one repository (Angad only — no vendored file touched, and `cbam-sync-check.mjs` still reports the vendored engine intact).

Commits	Repo	What
`31c4ba4` – `8936429`	Angad	Line model, plus two SHA-256 digests: a per-line fingerprint of inputs as entered, and a pack-snapshot hash replacing the `"browser-prototype"` placeholder
`5a98e98` – `fc7cf86`	Angad	Per-year threshold: lines grouped by calendar year and handed to `aggregateThresholdBasis` / `evaluateThreshold` ; `below_threshold` reachable only when the user ticks a per-year completeness attestation
`d434711` – `9b2120c`	Angad	CSV export: one row per line, engine values verbatim, each figure beside its legal locator
`800df31` , `9e8f641`	Angad	Renderers for the year-threshold card, running totals, per-line card and the printable audit document; the second commit fixed a defect that had printed a Commission workbook's hash under a regulation's label
`516fb3a` – `bdb8162`	Angad	Wired `lines[]` into the page: add/remove lines, export buttons, print stylesheet
`a700e22` , `f04b95b`	Angad	Post-ship fixes: Add-line failures now reported, focus restored on Remove, a stuck print class fixed, and an aria-live summary that had reintroduced the exact false certainty this branch exists to remove

The calculator went from pricing one line to pricing as many as the user adds, from a threshold verdict that could only ever say `indeterminate` to one that can say `below_threshold` on the user's own attestation, and from no export at all to a CSV and a printable audit document. Full detail in §4.3 and §4.4.

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2. The formulas

Two regulations, two halves of one calculation.

Certificates owed are proportional to:

```
(SEE  −  SEFA)  ×  mass  ×  CBAM factor  ×  certificate price
```

`SEE` is what the good emitted. `SEFA` is the deduction that keeps imports symmetric with EU producers still receiving free ETS allowances. Omit `SEFA` and you overcharge by the entire free allocation.

2.1 Embedded emissions — Annex IV, Reg (EU) 2023/956

```
Simple good:   SEEg = AttrEmg / ALg
Complex good:  SEEg = (AttrEmg + Σ mᵢ · SEEᵢ) / ALg
```

Our public calculator does not compute these. It reads the Commission's **default values** for the good, origin and production route, and applies the published mark-up. Computing `SEE` from plant data is the workspace product's job.

2.2 Free allocation — IR (EU) 2025/2620

```
Eq 1  FAAg      = SEFAg,y × Mg
Eq 2  SFAProcg  = CBAMy × CSCFy × BM*g          ← BM* is Column A
Eq 3  SEFAg     = SFAProcg                       (simple good)
Eq 4  SEFAg     = SFAProcg + Σ mᵢ · SEFAᵢ        (complex good)
Eq 6  SEFAg     = CBAMy × CSCFy × BMg            ← BM is Column B
```

The calculator runs **Equation 6 exclusively**.

2.3 Column A versus Column B

The single most consequential decision in the engine.

Column	Covers	Used when
A	this installation's own process step only	the emissions figure is process-only, with precursors declared separately
B	the whole finished good, precursors embedded	the emissions figure is all-inclusive

The column is chosen by the SCOPE of the emissions figure, never by whether the data was verified. An operator's verified actual data is normally all-inclusive and therefore still takes Column B.

Worked from our own rule pack — the cement chain:

CN	Good	Column A	Column B
25231000	Cement clinker, route (A)	0.666	0.666
25232900	Portland cement	**0**	0.666
25232100	White Portland cement	**0**	0.859

A clinker kiln calcines limestone, so its own process benchmark equals its whole-product benchmark. A cement grinder mills bought-in clinker and emits almost nothing itself, so its Column A is zero and its entire allocation arrives through the precursor term. The arithmetic closes exactly:

```
0 (process)  +  1 t × 0.666 (clinker's Column B)  =  0.666  =  cement's Column B
```

Column A plus Equation 4 is identical to Column B, done properly. Get it wrong on a default value and you deduct zero where 0.666 was owed — roughly **€4,890 per 100 t** of CN 25232900. Across the shipped pack, 102 of 141 goods publishing both columns disagree.

The Column A path is implemented, audited, and **deliberately unreachable** from the public tool. It is only correct against a verified process-only figure with a declared precursor list, and a screening tool can obtain neither.

3. Production routes

3.1 The vocabulary

Both regulations publish the same closed legend. Anything outside it is a bug.

Sector		
Cement	(A) grey clinker / cement	(B) white clinker / cement
Steel — carbon	(C) BF/BOF	(D) DRI/EAF · (E) Scrap/EAF
Steel — low alloy	(F) BF/BOF	(G) DRI/EAF · (H) Scrap/EAF
Steel — high alloy	(J) EAF	
Aluminium	(K) primary	(L) secondary

3.2 It is a matrix, not a list

The steel letters encode two orthogonal dimensions:

Grade ↓ / Process →	BF/BOF	DRI/EAF	Scrap/EAF
Carbon	(C)	(D)	(E)
Low alloy	(F)	(G)	(H)
High alloy	—	—	(J)

This is not written down anywhere in the regulations; it falls out of the legend. It explains everything below.

3.3 How a route is selected

```
country of origin
   └ IR 2025/2621 Annex I  →  "underlying production route determining CBAM BM"
        └ IR 2025/2620 §5.3 →  the benchmark row for that CN + route
```

The benchmark tables carry **no country column**. Country reaches the benchmark only through the route. Two rules travel with it:

- *"If no production route is indicated for a CN code, the CBAM benchmark is independent of the production route."* (2621 Annex I)
- Unlisted country, or a `–` in the cell, falls back to the "Other countries and territories" table.

Both are implemented.

3.4 The defect fixed in `397457e`

Annex §5.3 also publishes **production-year** variants: `(1)` for 2026-27 and `(2)` for 2028-30. These are validity periods, not routes.

`build-fa-package.py` was writing them into the `routeIndicator` field and giving every row one open-ended validity. Two consequences:

1. Both variants stayed active for every date, so the date could not choose.
2. Nothing ever matched them — the defaults corpus never declares a route of `(1)`, nor a compound `(F)(1)`.

794 of 2,465 rows were unreachable.

The fix moves the year into `validFrom` / `validTo`, where `resolveBenchmark` already looks — it filters `active(validFrom, validTo, date)` *before* matching the route. No engine code changed. The consumer was right; only the producer was wrong.

IR 2025/2621's own legend, read afterwards, lists (A) through (L) and contains no `(1)` or `(2)` — independent confirmation from the sister regulation.

Route distribution in the live pack, post-fix:

```
(none) 800   (C) 309  (D) 309  (E) 309
(F) 151  (G) 151  (H) 151  (J) 151
(K) 63   (L) 63   (A) 4    (B) 4
```

Zero year markers. `(F)` – `(J)` at 151 each are the former standalone rows plus their two former year-variants, now separated by window instead of by name.

3.5 Combined routes — known, not yet resolved

IR 2025/2621 Annex I publishes `(C)/(F)` **140 times** and `(E)/(H)` **25 times**. Read against the matrix, these are *process known, grade ambiguous* — the only combinations that exist are process-matched pairs. `(C)/(D)` and `(D)/(G)` never appear.

For every such CN the 2620 table publishes exactly **one** of the two halves, because the CN description settles the grade (§5.2.3: "*low alloy steel means alloy steel other than high-alloy*"). Checked across all 19 affected goods: one half present, zero ambiguous, zero missing.

`resolveBenchmark` does exact string equality, so `(C)/(F)` matches nothing and those goods fail closed. Splitting on `/` and taking the published half would recover 19 of the remaining 23. **Not yet implemented.**

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4. The tool

4.1 Inputs

Good (CN code) · Country of origin · Production route · Emissions scope · Net mass (tonnes) · Import date.

The route list is never free-typed — it is derived from the routes the defaults corpus actually publishes for that CN and origin. The emissions-scope control appears only for goods the Commission publishes an indirect default for (cement, fertilisers, sintered iron ore); elsewhere it cannot change the answer.

4.2 What it returns

Status	Meaning
`cscf_pending`	A labelled what-if. No final figure exists.
`unavailable`	Fails closed — names the missing rule selector verbatim.
`zero_by_fiat`	Electricity: nil allocation by law (Art 1(2)), not by calculation.

94.3% of all resolvable answers are `cscf_pending`. Zero are settled figures. The cross-sectoral correction factor is unpublished for 2026-2030 and is **not 1.0 by default**; the engine refuses to assume one and shows a labelled scenario at the last value the Commission actually set — `CSCF_2021_25 = 1` (`sefa.ts`).

That scenario is a floor, not a midpoint. The CSCF only ever scales the free allocation deduction (SEFA) down from its benchmark ceiling — it is bounded above by 1, never above it — so pinning it at 1 computes the *largest* deduction the law can produce and therefore the *smallest* certificate figure the good can owe. The real CSCF for 2026-2030, once published, can only be equal to or lower than 1. **A displayed figure cannot fall; it can only rise or hold.** Since 94.3% of answers are what-ifs, this governs almost every number the tool produces — an importer who reads a `cscf_pending` figure as a ceiling, rather than a floor, will under-provision.

4.3 The de minimis threshold

50 t per calendar year, across cement, aluminium, fertilisers and iron & steel (Reg 2023/956 Art 2(3)).

```
above 50 t                           → above_threshold
at or below + completeness=complete → below_threshold
at or below + partial/unknown       → indeterminate
```

"**Above**" **can be proven from one line.** "**Below**" **cannot** — it requires the whole year. The calculator now takes multiple lines and evaluates the threshold **per calendar year** through `aggregateThresholdBasis`; `below_threshold` is reachable only when the user explicitly attests the list is their complete year, and every surface that shows the verdict names that attestation as its basis. Estimates export as a CSV (engine values verbatim, one row per line, each beside its legal locator) and as a printable document whose final section states what the figures cannot tell you.

4.4 Export: CSV and the printable document

CSV. One row per line, engine values verbatim — no locale formatting, no rounding. The identity that reconciles a row is not "chargeable equals embedded minus free allocation" — that fails on two counts the regulation itself requires:

```
chargeable_tco2e = max(0, direct_tco2e − free_allocation_tco2e) + indirect_tco2e
```

Free allocation is a direct-emission benchmark (SEFA, §2.2), so it is deducted from the direct side only — indirect is added back **after** the deduction, not folded into what gets deducted from. And the deduction floors at zero: the regulation surrenders certificates, it never issues them, so a clean producer whose free allocation exceeds its own direct emissions cannot generate a negative charge.

Both halves are reachable with ordinary catalogue goods, checked against the live pack:

CN / origin / route	direct	free allocation	indirect	chargeable	What it shows
`25232900` / AL / default, 100 t	99.0	64.935	3.3	37.365	Deduction clamps nothing; indirect still adds on top
`25070080` / AO / (A), 100 t	24.2	64.935	4.4	4.4	Free allocation exceeds direct emissions; the deduction floors at 0 and the whole bill is the indirect component

Printable document. Four sections — what you asked, what we computed, on what authority, and what this does not tell you. The last section is the differentiator: it states the CSCF is unpublished and every figure a floor (§4.2), that Art 9 carbon-price deductions are not modelled, that any below-threshold verdict rests on the user's own completeness statement, and that the line fingerprint covers inputs as typed, never a source document.

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5. Data and provenance

Live pack, `generatedAt 2026-08-07T16:59:36.563Z`:

Table	Rows
`defaultFactors`	41,100
`benchmarks`	2,465 (Column A 661, Column B 1,804)
`classifications`	574
`cbamFactors`	9 — 2026 is **0.975**, free allocation *retained*
`cscf`	5 — all `pending`, 2026-2030
`prices`	4 — 2026-Q1 €75.36, Q2 €75.28, Q3/Q4 unpublished
`thresholds`	1 — 50 t

7.55 MB raw, **281 KB on the wire** (Vercel serves brotli). Repeat visits transfer nothing.

5.1 Source hashes

Two regulations now carry real hashes, replacing sixty-four zeros:

```
IR 2025/2620   8bbba79e7f33f0e4943140c28e91a8810612f2fa770bd6dcad33fdb7045e4c05
IR 2025/2621   3155016c2e07b049b64f1ac4c2320061534245b81971ce5cba7814736f09acb4
```

Both PDFs were retrieved by hand and verified as genuine Publications Office artefacts. EUR-Lex sits behind an AWS WAF challenge, so neither can be re-fetched by a script — the hash pins *which* text we read; it cannot be re-verified automatically.

Four sources still carry placeholders: `dir-2003-87-art-10a-1a`, `dr-2019-331-art-14-6`, `reg-2023-956`, `ec-certificate-price-page`.

5.2 The second defect, fixed in the same commit

`build-fa-package.py` could not reproduce its own golden file. The `thresholds` block — the 50 t gate — and its `reg-2023-956` source had been hand-added on 2026-07-29 and never taught to the generator. **Every regeneration silently deleted the rule deciding whether an importer owes anything at all.**

Invisible until something regenerated. The year-window work forced a regeneration, which is how it surfaced.

The near-miss is worth recording: the first fidelity check compared `benchmarks` only, and passed. A full-package comparison found **two** drifts — `thresholds` *and* the `sources` array. **A fidelity check must compare the whole artefact; comparing the part you are about to change proves nothing about the parts you are not.**

5.3 Three digests, three different claims

The export (§4.4) and this document both now carry real hashes where the tool used to carry a placeholder — but they are not one claim wearing three names, and conflating them was a real defect caught in review before this branch shipped:

Digest	Over	Says
IR 2025/2620, IR 2025/2621 (§5.1)	the enacted regulation PDFs themselves	this is the binding legal text we read
Commission benchmarks / default-values workbooks	the Commission's own Excel workbooks	this is the informational transcription the defaults corpus was built from — not the binding text
`pack_snapshot` (CSV column) / `stamp.snapshotHash`	`generatedAt` plus both workbook hashes, SHA-256'd together	this is the exact corpus a given figure was computed from

```
Benchmarks workbook
b79108b025e697822f0f59de477fa68066c1c05c228fae2270cd230af84e8a7b
Default-values workbook
865372ed23649b7b02c9124f207fc0b0875fd244c45c19e9fb8cdb1e503a5003
```

An earlier build of the printable document read the workbook hashes but labelled them "IR (EU) 2025/2620" and "IR (EU) 2025/2621" — every character of each hash was real, but printed under the wrong artefact's name, which is a false provenance claim regardless. Fixed before ship (`9e8f641`): the document now prints the regulations' own hashes under those labels and the two workbook hashes separately, labelled as transcriptions. `stamp.snapshotHash` no longer reads `"browser-prototype"` ; every estimate is decorated with the real pack-snapshot hash before it renders.

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6. How it is verified

- Unmodified generators reproduce their committed goldens **byte-for-byte**, before any change is made.
- A migration gate asserts every benchmark value, every `sourceLocator` and every non-benchmark section is unchanged.
- A regression test asserts no year marker can reappear in `routeIndicator` , and no `(CN, column, route)` group holds both a bounded and an unbounded row.
- `cbam-sync-check.mjs` hashes the eleven vendored engine files against a committed manifest. The engine in the website is a copy of the SaaS's, and this is the tripwire against drift.
- CBM: 379/379 tests, typecheck clean. Angad: 171/171 unit, 16/16 Playwright, publication contract 80 checks / 0 violations.

CBM's own `differential.test.ts` had pinned this gap at 382 priceable / 183 stranded for India and concluded it "*needs corpus research rather than a code change.*" It was a code change. The pin now reads 528/37, and the remaining 37 are a genuine corpus question.

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7. Open

Item	Note
CSCF unpublished	94.3% of answers are what-ifs. Nothing we can do; a one-line data change when Brussels publishes.
2027+ import dates	Certificate prices exist only for 2026 quarters, so any later date fails closed with no explanation. The most visible remaining flaw.
Combined `(C)/(F)` routes	19 goods, resolution backed by both regulations. Not implemented.
Art 9 carbon price paid abroad	Not modelled; disclosed in output as making the figure conservative. Implementing act still a draft.
`Art 2(2)` citation	Wrong in five places in the vendored engine — Article 2 has no numbered paragraphs; the provision is Art 1(2). Unreachable today. Our own new `cscf_status` CSV column, which prints this same fact for electricity's `zero_by_fiat` rows, cites it correctly as Art 1(2) — the two do not contradict each other; one is our export, the other is the still-unfixed vendored `sefa.ts`.
41,100 default factors	Never reconciled against IR 2025/2621's 2,400-page Annex I. The largest unaudited surface.

Sources: IR (EU) 2025/2620 and 2025/2621 (OJ PDFs, hashes above); Reg (EU) 2023/956; the shipped rule pack at deltaclimate.earth. Benchmark figures quoted are from our own pack, cross-checked against the published Annex.